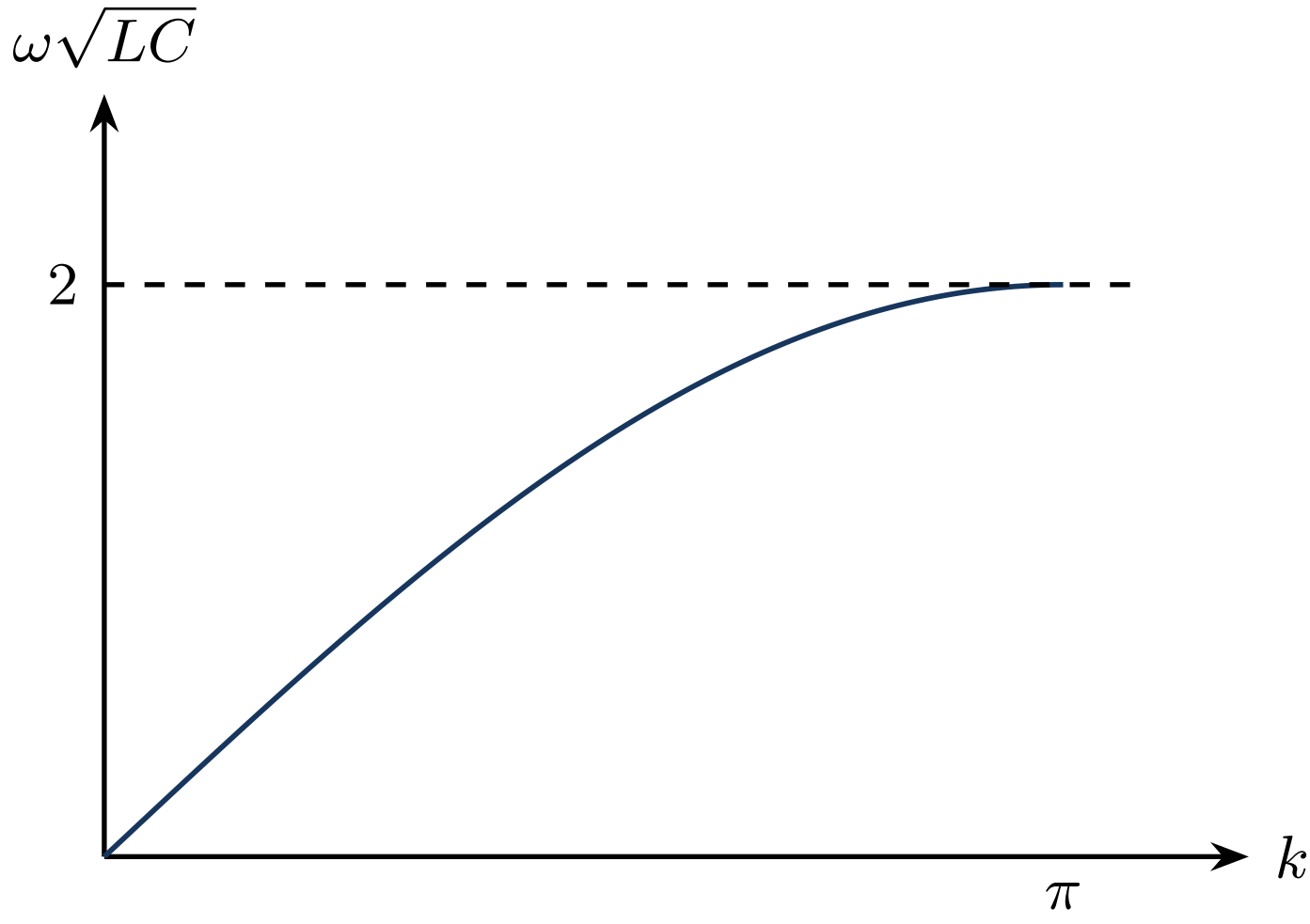

$\omega\sqrt{LC} = 2\sin(k/2)$；带边群速度趋零